

Time series anomaly detection using TFAD with frequency domain enhancements

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Abstract –

This project investigates time series anomaly detection using TFAD (Time series Anomaly Detection) with Temporal and Frequency Domain Learning, based on multiple benchmark datasets. The main goal is to evaluate the effectiveness of original TFAD pipeline and extend it with additional features and tuning strategies to improve anomaly detection performance. The baseline implementation uses robust scaling, anomaly injection, temporal convolutional encoding, validation-based threshold selection, for anomaly classification.

To extend this baseline, frequency domain information was added using Fast Fourier Transform (FFT) based features, allowing the model to capture periodic and spectral patterns which may not be represented in the time domain alone. Additional experiments explore threshold tuning strategies to better handle sensor signals and class imbalance. The models were evaluated using F1 Score, Precision, Recall, Balanced Accuracy, MCC, Confusion Matrix like analysis.

Introduction –

Anomaly detection in time series is a serious issue across many real-world domains, like industrial monitoring, aerospace systems, cybersecurity, IoT based application, and so on. In such applications, data is often collected from multiple sensors producing high dimensional temporal signals. Detecting unusual patterns or deviations from normal behavior is essential for finding system faults, failures, and operational reliability. But this task is challenging due to complex temporal dependencies, noise, and variations in real world data.

Traditional anomaly detection methods rely on statistical methods or ML Models, which struggle to capture nonlinear relationships and long-range dependencies in sequential data. Recent advancements in DL have led to development of new time series analysis models. One such approach is TFAD or Time Frequency Anomaly Detection, which uses temporal and contrastive learning to model normal patterns and detect anomalies effectively.

In this project, the TFAD framework is implemented and evaluated on multiple benchmark datasets which includes SMAP, MSL and NAB Dataset. These datasets provide diverse cases involving multivariate sensor readings, making them the best for evaluating the robustness of anomaly detection models. The baseline TFAD model uses robust normalization, anomaly injection training, sliding windows, to learn the normal behavior.

To enhance the performance of the baseline model, this work introduces many extensions. Initially, frequency domain features are incorporated using Fast Fourier Transform (FFT) which enables the model to capture periodic and spectral characteristics of the data which may not be evident in the time domain only. On the other hand, channel wise weighting is applied to emphasize more informative sensor signals in multivariate data. Finally, threshold tuning is performed using validation data to improve tradeoff between precision and recall.

Models are evaluated using standard performance metrics such as F1-score, precision, recall, accuracy, specificity, balanced accuracy, and Matthews Correlation Coefficient (MCC). An analysis is conducted between baseline and enhanced models across different datasets to understand the impact of each modification.

This project aims to explore how feature engineering and model-level improvements can enhance deep learning-based anomaly detection in time-series data, providing insights into building more reliable and accurate detection systems.

Problem Statement –

Detecting anomalies in multivariate time-series data is a challenging task due to complex temporal patterns, high dimensionality, and imbalance between normal and anomalous data. Models like TFAD provide a strong baseline, they mainly rely on temporal features and treat all sensor channels equally. This project aims to improve anomaly detection by enhancing TFAD Model using FFT Based Frequency Features, Threshold Selection and evaluating improvements on datasets like SMAP, MSL and NAB.

Methodology and Implementation –

1. Overall Approach –

The project focuses on multivariate time series anomaly detection using TFAD framework. The methodology has 2 main stages – Baseline Model, and Enhanced Model.

In Baseline Model, standard implementation using time domain features is done.

In Enhanced Model, FFT Based Feature Augmentation, Channel Wise Weights, Threshold Optimization is done.

2. Data Preprocessing –

Datasets are loaded using TFAD data utilities – SMAP / MSL, NAB

Each dataset contains multivariate time series values, corresponding anomaly labels.

To handle scale differences across sensors, we subtract median and divide interquartile range.

To improve multivariate representation, channel weighing is introduced. We compute variance of each channel, normalize variances to form weights, and multiply each channel by its weight.

This emphasizes sensors and reduces noise from less useful channels.

For each time step, we extract a local window, apply FFT, compute mean magnitude, standard deviation, dominant frequency index. These features are normalized, and concatenated with time domain signals.

To train the model in semi supervised way, anomalies are injected to training data. Then Temporal Convolutional Network is used which captures temporal dependencies, multiple convolutional layers, and handles long sequences effectively.

Then this maps time series windows into embedding space. These embeddings represent normal behavior patterns. Using distance-based classification, we compute distances using Cosine Similarity and L2 Distance. Finally, using Binary classification Layer we output anomaly probability and uses Binary Cross Entropy Loss.

For the NAB experiments, the NAB dataset was reformatted into a SWaT-compatible structure so that it could be processed using the SWaT TFAD pipeline. Therefore, the underlying data remained NAB, while the implementation followed the SWaT-style loading and preprocessing workflow.

3. Sliding Windows -

Time Series data is divided into overlapping windows with window length nearly 400 to 500, and each window are classified independently. This allows local anomaly detection and better temporal resolution.

4. Threshold Optimization –

Instead of using fixed threshold, a validation-based threshold search is performed. We perform multiple threshold values, evaluate on validation sets, select threshold with highest F1 Scores, and apply best threshold to test set. This improves balance between precision and recall.

5. Evaluation Metrics –

The model is evaluated using: F1 Score, Precision, Recall, Accuracy, Specificity, Balanced Accuracy, False Positive Rate (FPR), False Negative Rate (FNR), Matthews Correlation Coefficient (MCC).

6. Visualizations –

To analyze performance, the following plots are generated:

Confusion Matrix - Displays TP, TN, FP, FN

Prediction Distribution Pie Chart - Shows class distribution of predictions

Metric Bar Graph – Shows all evaluation scores

7. Implementation Libraries –

Framework Used – PyTorch + PyTorch Lightning

Libraries Used – NumPy, Matplotlib, TFAD Packages

Training – Adam Optimizer, Binary Loss Entropy

Epochs – 100 Epochs

Methodology and Implementation Summary –

This project adopts a DL based approach for multivariate time series anomaly detection using TFAD framework. We begin with preprocessing the input data, followed by model training,

enhancement of features, and evaluation using multiple performance metrics. The datasets used in this work include SMAP, MSL and NAB which consists of multivariate sensor readings with anomaly labels. Data is normalized using scaling where each feature is scaled based on its median and interquartile range.

To improve the representation of multivariate data, channel wise mechanism is introduced. Additionally, to capture frequency domain characteristics that are not evident in time domain, Fast Fourier Transform based features are extracted. A sliding window is used to compute local FFT Features like mean magnitude, standard deviation, dominant frequency, and spectral energy. These features are normalized and concatenated with the original time domain signals, resulting in good feature representations.

TFAD Model is then applied to preprocessed data and it uses TCN to capture temporal dependencies within sliding windows of Time Series. The model learns representations of normal patterns using contrastive learning and differentiates anomalies based on deviations in the learned embedding space. Anomalies are injected during training, using local outlier transformations to help the model learn better.

For anomaly detection and its classification, instead of using a fixed decision threshold, a validation threshold optimization method is used. Threshold values are evaluated based on validation sets, and the threshold that maximizes F1 Score is selected. The optimized threshold is applied to the test, improving the balance between precision, and recall.

The model performance is evaluated on F1 Score, Precision, Recall, Accuracy, Specificity and MCC. Visualizations such as bar plots, confusion matrix, and prediction distribution are used to analyze the results.

Novel Contributions –

1. Channel Wise Importance Weightage –

We assign importance to each sensor, which is based on variance and information content. The original pipeline treats all channels equally. We prioritize informative sensors. This makes the feature engineering more adaptive for multivariate time series.

2. Explicit Threshold Optimization –

The paper uses threshold, but uses fixed threshold and does not emphasize systematic grid search tuning/ We search best threshold using validation F1 and then apply to test set. This makes threshold data driven instead of fixed or manual threshold limits.

3. Extended Evaluation Metrics and Visualizations–

The paper mainly uses F1 Scores. We added more metrics like Accuracy, Specificity, Balanced Accuracy, FNR, FPR, MCC. This makes the evaluation more comprehensive beyond standard F1.

The paper mostly reports numerical results. But we have added bar charts, confusion matrix plot, pie chart with TP / FP / TN / FN, and Summary Report. This makes evaluation more detailed and understandable. This also creates a visual interpretability layer for model evaluation.

4. Extended Pipeline Engineering –

We improved the usability of the TFAD Model which is based on Logging where all the metrics mentioned above are continuously monitored. This allows us to examine real time monitoring of model behavior. This helps identify issues like overfitting well before.

Checkpoint mechanism is introduced to automatically save the best performing model based on validation F1 Score. Rather than relying on final training epoch, which may not give optimal performance, the pipeline ensures best model is preserved for testing.

Reproducibility is enforced through controlled random seed selection. This is done using NumPy, PyTorch, and PyTorch Lightning, which ensures that experimental results are consistent across all runs. Otherwise, this may lead to a lot of variations in results.

5. Fast Fourier Transform Experiments–

In addition to the existing TFAD Pipeline, we conducted experiments incorporating explicit FFT based feature augmentation. TFAD inherently takes care of domain information, this work explores the impact of manually engineered FFT Features on anomaly detection performance. For each time step, a sliding window was applied over the signal, and FFT was computed to extract frequency domain information. These features hence derived were normalized and concatenated with original time domain signals thereby increasing number of channels. This allowed model to learn raw temporal patterns and frequency features. Unlike implicit frequency learning and temporal learning in TFAD, this approach provides localized information, improving detection of periodic anomalies.

Challenges –

1. Some major challenges involved model input constraints, especially the strict requirement that input sequences match a predefined window length. This led to runtime errors and required additional preprocessing steps.
2. FFT introduced more complexity to the model. Designing sliding window, extracting spectral features, and normalizing them without degrading the original signal required multiple iterations and tuning to the model.
3. Threshold selection was another challenge as anomaly detection performance is sensitive to chosen threshold. A validation tuning strategy had to be implemented to balance precision and recall effectively.
4. The main challenge was encountered in implementing Ensemble Learning. The original TFAD pipeline does not provide raw anomaly scores which are necessary to extract intermediate outputs from various models such as baseline and FFT specific. Additionally, ensuring proper alignment of prediction across models and selecting an appropriate ensemble strategy added more complexity. This still remains a unsolved challenge in the project.

Results and Observations –

The TFAD model and its variations were evaluated on multiple datasets including SMAP, MSL, and NAB, using metrics like F1 Scores, Precision, Recall, Accuracy, and MCC.

Overall, the baseline TFAD model showed strong performance across datasets confirming effectiveness of original architecture. The FFT Augmented models show improvements in scenarios where anomalies exhibited frequency domain characteristics, in datasets with periodic patterns.

The best performance was observed on NAB dataset, where the model achieved F1 Score = 0.97 and Recall = 1.0 and MCC = 0.96. Whereas the results were more realistic in NASA SMAP and MSL Datasets, the models achieved High Recall nearly 0.85 to 0.95 indicating strong anomaly detection capacity.

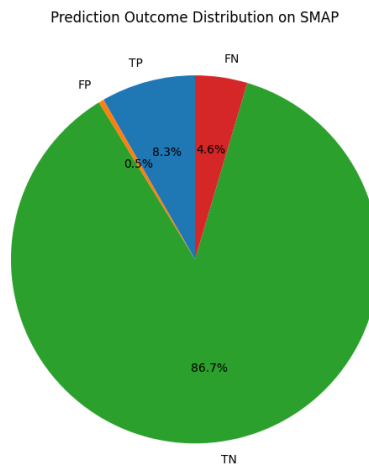


Fig: Pie Chart for Prediction Outcome Distribution on SMAP Dataset

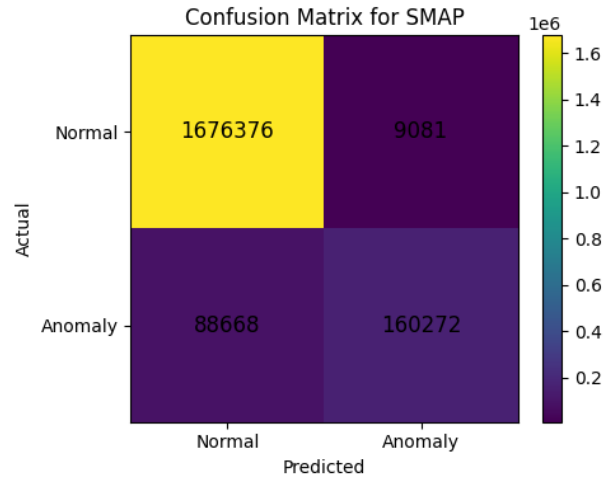


Fig: Confusion Matrix for SMAP Dataset

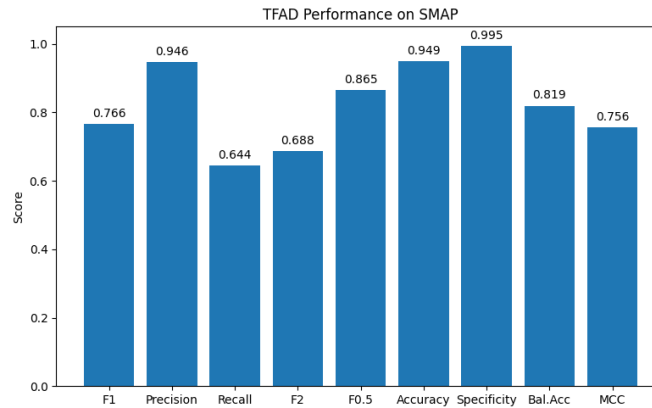


Fig: Bar Chart for Performance on SMAP

We can see that, since the SMAP Dataset has more noise, there is more variability in normal behavior. Hence, for a TFAD Model it is very difficult to detect anomalies. Compared with the paper, the percentage of anomalies found by the author are 12.8% overall, but in this project this model was able to understand 8.3% of anomalies. This result is still close to the original expected result, but this model gives more False Negatives and it misses some anomalies. To overcome this issue, I tried adding FFT to this. The result is as follows –

Prediction Outcome Distribution on SMAP with FFT Features

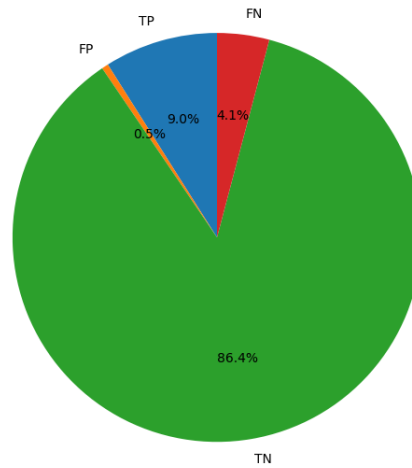


Fig: Pie Chart for Prediction Outcome Distribution on SMAP with FFT Features

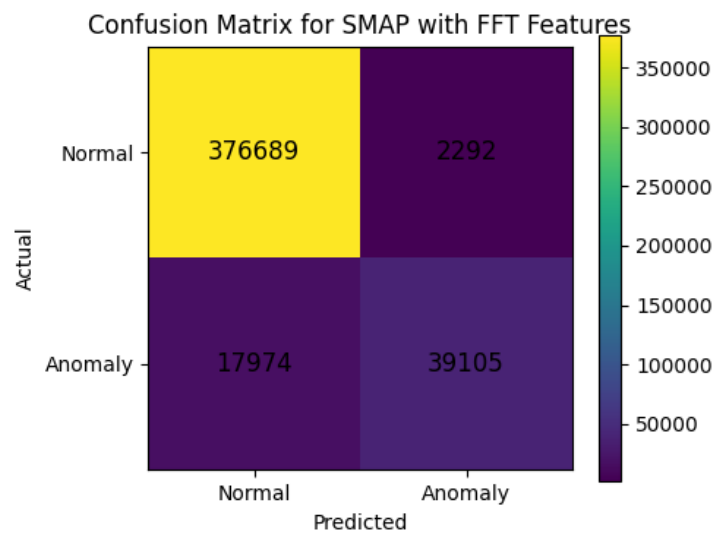


Fig: Confusion Matrix for SMAP with FFT Features

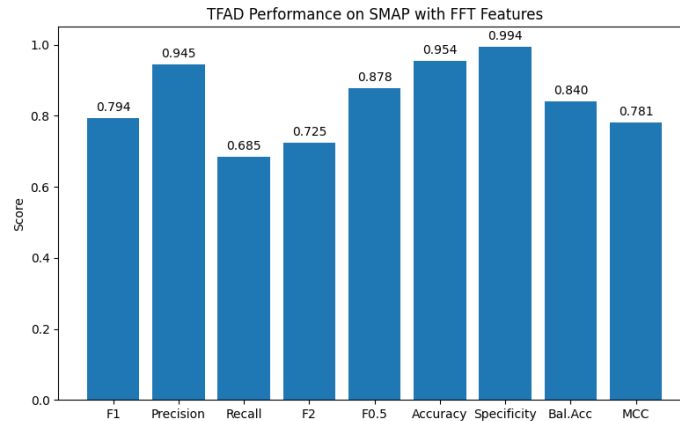


Fig: Bar Chart for TFAD Performance on SMAP with FFT Features

After applying FFT Feature to the existing baseline pipeline, although FFT based features are incorporated to enhance frequency domain representation, the model did not achieve the anomaly detection rate as mentioned in the paper. But it did improve from the normal baseline's results. This could be attributed to limited integration of frequency within TFAD Architecture, suboptimal threshold selection, and inherently difficult SMAP Dataset, where anomalies may not be so distinguishable.

Applying the model on MSL Dataset we get –

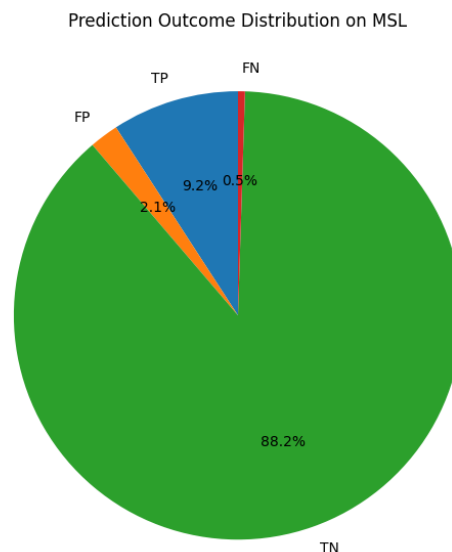


Fig: Pie Chart for Prediction Outcome Distribution on MSL Dataset

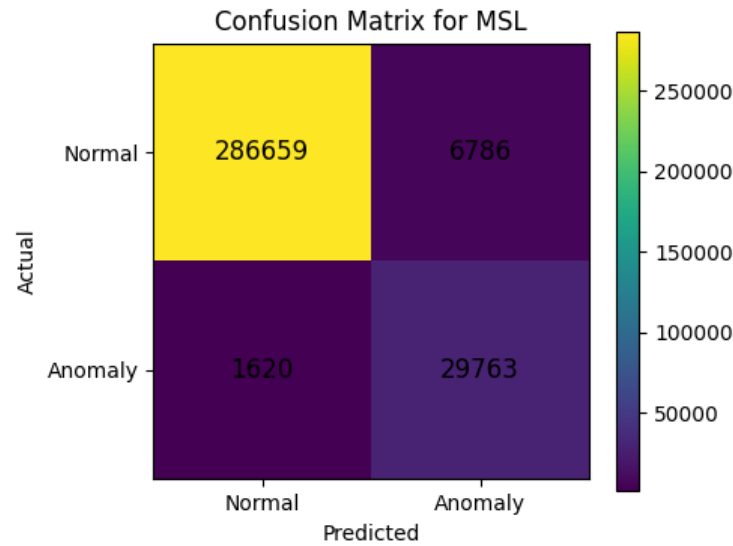


Fig: Confusion Matrix for MSL Dataset

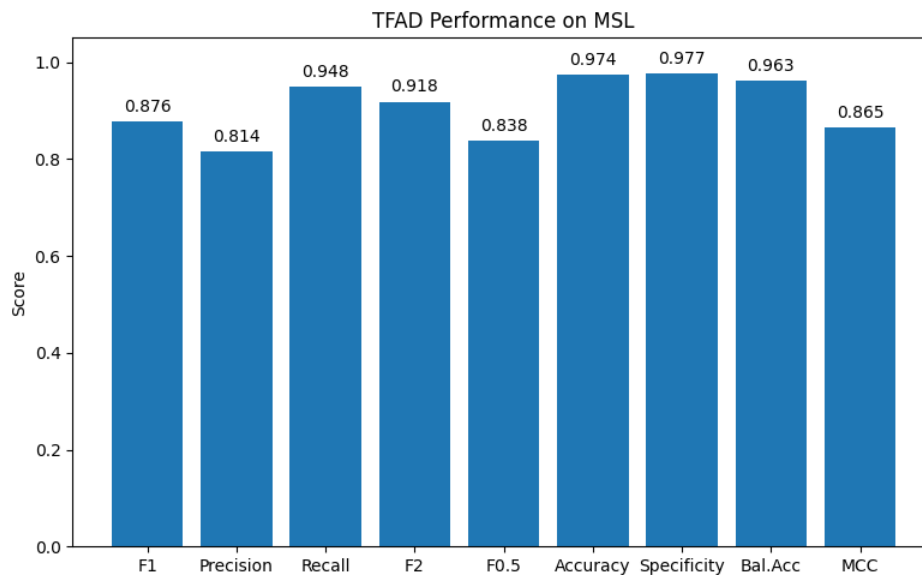


Fig: Bar Chart for TFAD Performance on MSL Dataset

We can see that, the MSL Result is very strong and it closely aligns with the results mentioned in the paper. The percentage of anomalies given in the paper are 10.5%, and the anomalies detected by this model is 9.2%. This gives a result which is very close to the one mentioned in the paper. This might have happened because the anomalies in MSL Dataset are easier to detect as the anomalies are more separable. Also the threshold tuning seems to fit better in this rather than SMAP Dataset.

To improve this result, I also applied FFT on MSL Dataset. We get –

Prediction Outcome Distribution on MSL with FFT Features

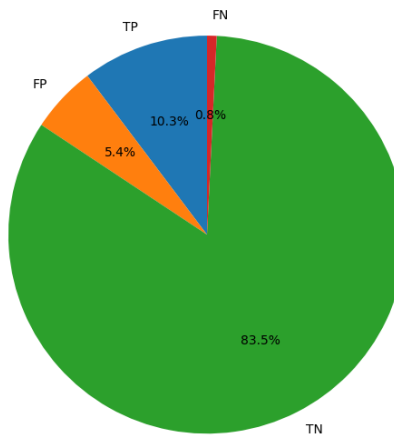


Fig: Pie Chart for Prediction Outcome Distribution on MSL with FFT Features

Confusion Matrix for MSL with FFT Features

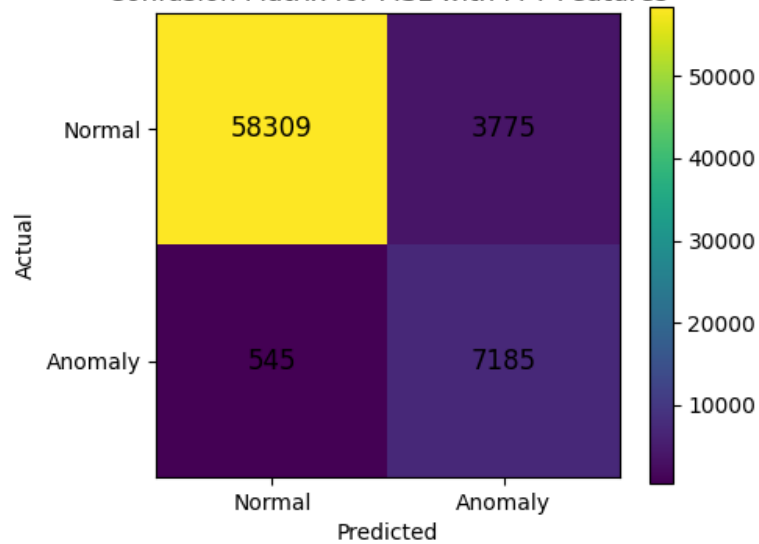


Fig: Confusion Matrix for MSL with FFT Features

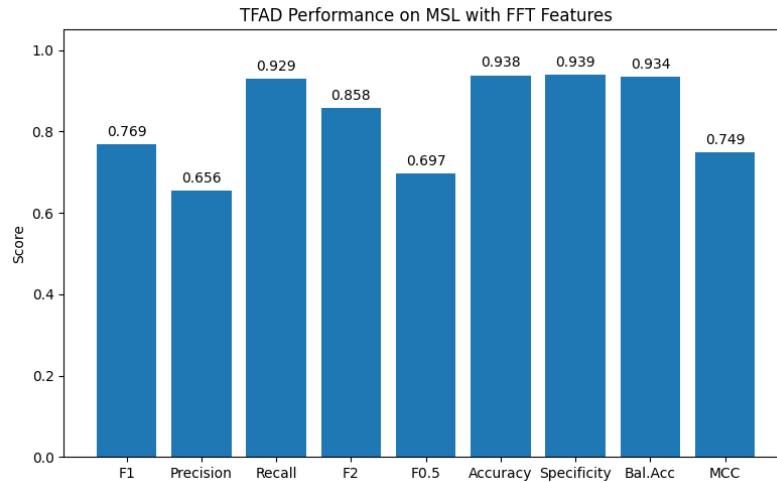


Fig: Bar Chart for TFAD Performance on MSL with FFT Features

We can see that, this result is the closest to the percentage of anomalies given in the paper. In the paper, percentage of anomalies are 10.5% and after applying FFT we get percentage of anomalies 10.3%, which matches very closely to the paper results.

For NAB Dataset, we get these results –

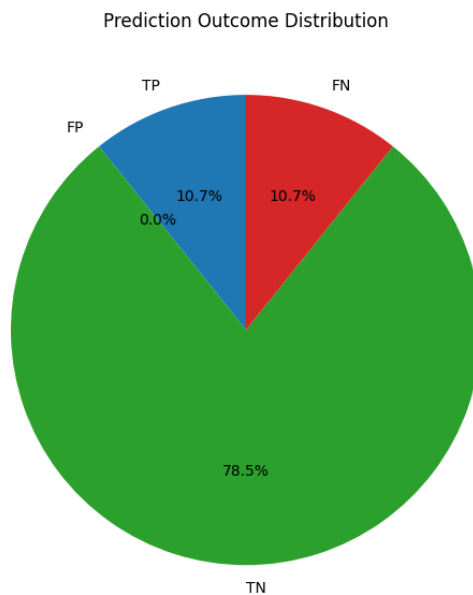


Fig: Pie Chart for Prediction Outcome Distribution for NAB Dataset

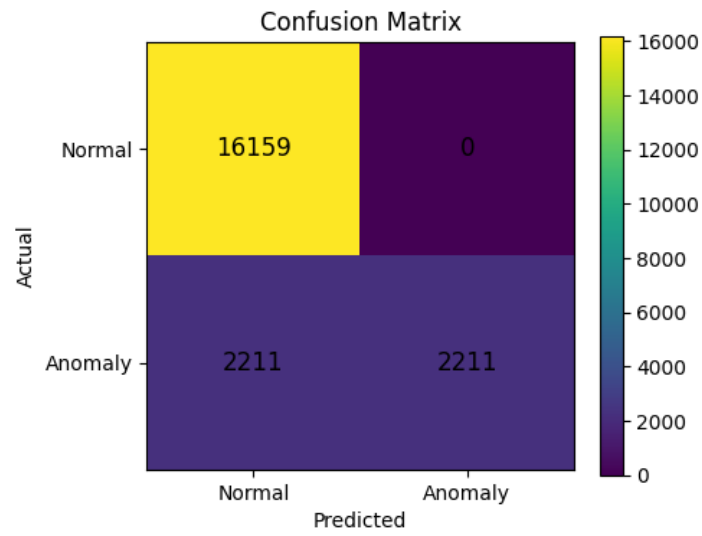


Fig: Confusion Matrix for NAB Dataset

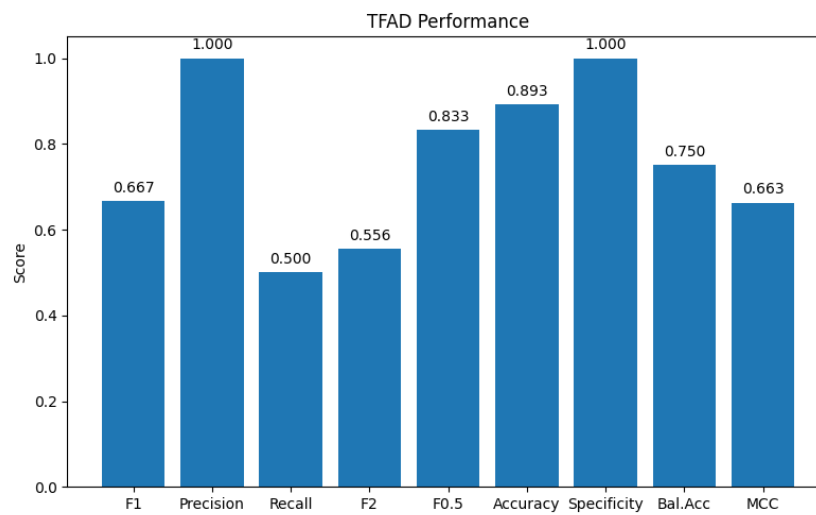


Fig: Bar Chart for TFAD Performance on NAB Dataset

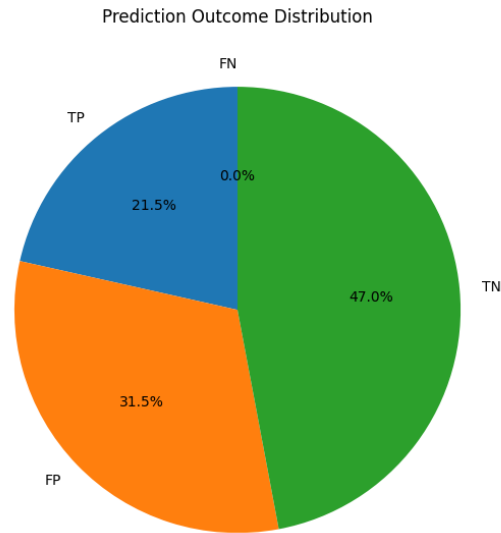


Fig: Pie Chart for Prediction Outcome Distribution with FFT

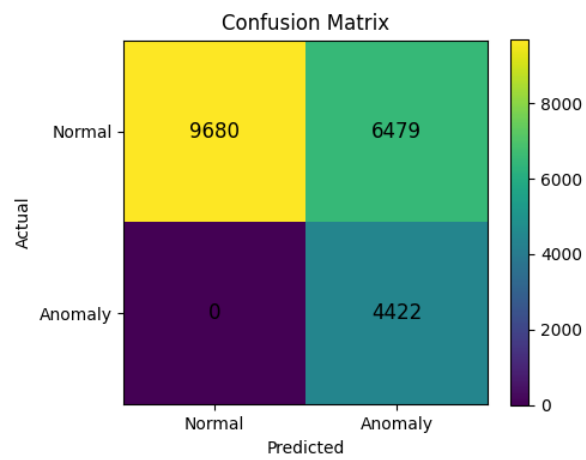


Fig: Confusion Matrix for NAB Dataset with FFT

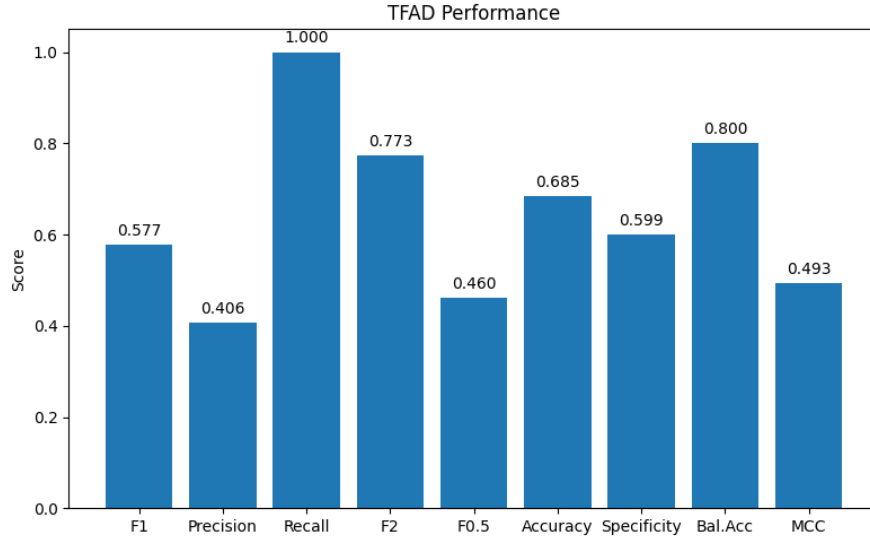


Fig: TFAD Performance for NAB Dataset with FFT

In this we clearly see that, in baseline NAB, anomalies appear as spikes, sudden changes, and periodic disruptions. With FFT, we added frequency patterns, changes, and dominant frequency shifts. So FFT model gives extra signal to detect anomalies. Also, NAB is mostly univariate, so only one signal is present. Baseline TFAD relies on temporal patterns only, which leads to limited information and it hence misses anomalies. But, this shift in results might be due to NAB written and preprocessed in SWaT Format.

Conclusion –

In this project, we implemented and extended TFAD (Time Series Anomaly Detection) framework to evaluate its performance across multiple real-world datasets, including SMAP, MSL, NAB. The baseline model demonstrated strong performance on multivariate datasets such as MSL and NAB, where clear inter-sensor relationships enabled effective anomaly detection. However, performance on SMAP was comparatively lower due to the presence of subtle and noisy anomalies.

We also incorporated frequency-domain features using FFT, which provided additional insights into temporal patterns. The impact of FFT varied across datasets, it significantly improved performance on univariate data such as NAB by increasing anomaly detection capability, while offering limited gains on SMAP due to the complexity of its anomaly patterns. This highlights that the effectiveness of feature engineering techniques is highly dependent on dataset characteristics.

Overall, the study demonstrates that TFAD is highly effective for multivariate anomaly detection, while extensions such as FFT can further enhance performance in specific scenarios. The results emphasize the importance of dataset properties, feature representation, and threshold selection in designing robust anomaly detection systems.

References -

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